PH-EVAL v0.1: a descriptive comparison of five open instruct models on native Filipino and Taglish tasks

PinoAI research note · EVAL-001 · descriptive

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! Correction (2026-09-30)

After publication, a scoring bug was found: the step that strips a leaked "thought" marker from Gemma 4 12B's answers missed two of its four forms. With the fix, **Gemma 4 12B scores 63/70 (90.0%) on the automatically scored subset, not 61/70 (87.1%)**. No other model is affected, and the ranking is unchanged. In the paired Gemma-vs-Qwen comparison on this subset, the result becomes 9:0 ($p = 0.0039$), so the statement below that this subset alone could not separate the two models no longer holds. The text and tables below are the original analysis, kept for the record.

Truncation (2026-09-30). The truncation counts below (12 to 24 of 175) came from a text-length proxy that undercounted badly. Measured with each model's own tokenizer, 30–37% of every model's answers reached the 512-token cap: Gemma 4 12B 53, Qwen3.5-9B 64, Sailor2-8B-Chat 59, Qwen3.5-4B 65 and Gemma 4 E4B 54 of 175. All five models are being re-run with a 4,096-token cap before any further comparison.

Abstract

We ran PH-EVAL v0.1, a native-authored Filipino and Taglish evaluation of 175 items, on five open-weight instruct models: Gemma 4 12B, Gemma 4 E4B, Qwen3.5-9B, Qwen3.5-4B and Sailor2-8B-Chat. All five used the same prompts, the same code path, thinking disabled and a 512-token cap. This note is **descriptive**. It tests no hypothesis.

- **Automatically scored subset (70 items):** Gemma 4 12B 87.1%, Gemma 4 E4B 80.0%, Qwen3.5-9B 77.1%, Qwen3.5-4B 68.6%, Sailor2-8B-Chat 61.4%. Adjacent models' 95% intervals overlap widely.
- **Free-generation items (105):** an LLM judge settled only 16–39 per model, all as confident fails. It routed the other 66–89 to a human.
- **Language drift:** none. The median Tagalog function-word share of free-generation answers is 0.97–1.00 for every model.

The pre-registered, human-rated paired comparison of three of these models (Gemma 4 12B, Qwen3.5-9B and Sailor2-8B-Chat) is reported separately in RESULT-001. Here we only describe all five side by side.

1. Introduction

PinoAI needs a base model to adapt to Filipino and Taglish, and a benchmark to track progress. PH-EVAL v0.1 is that benchmark's first version. A pre-registered test on three candidates is already published (RESULT-001). Two more models were run through the identical pipeline afterwards:

- **Qwen3.5-4B**, a smaller sibling of Qwen3.5-9B;
- **Gemma 4 E4B**, the smaller Gemma 4 model and the registered fallback if Gemma 4 12B proves too expensive to train.

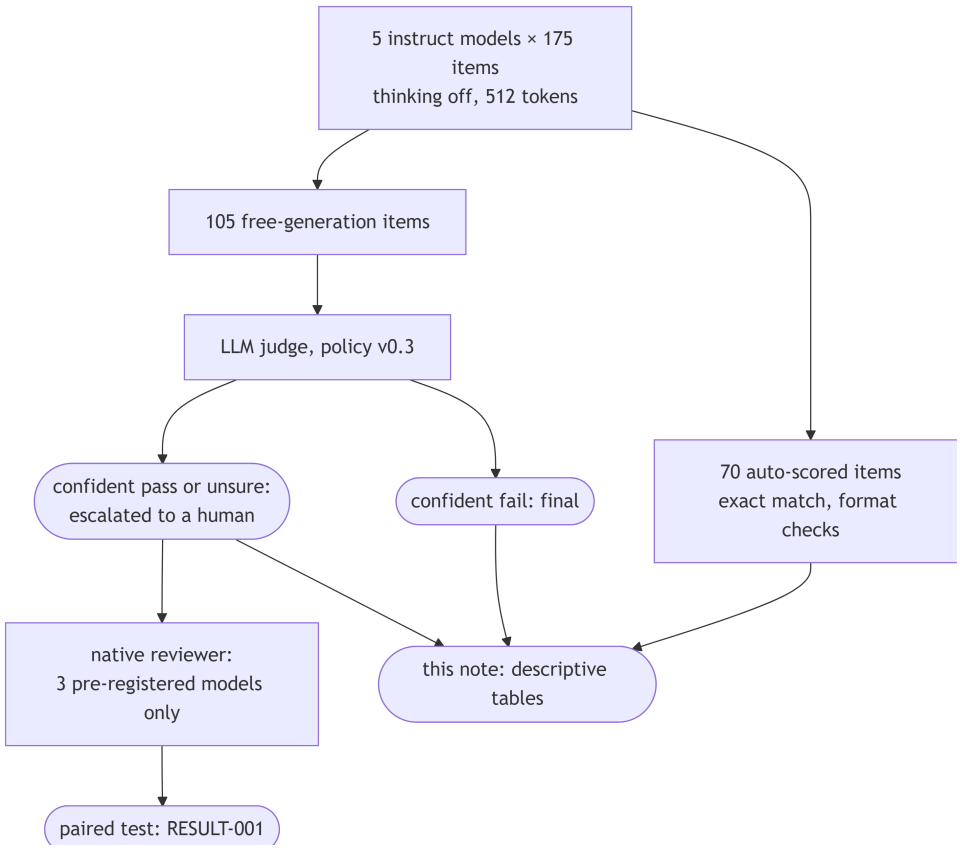
This note puts all five in one place, so the benchmark's behaviour and the models' relative standing can be read without over-reading them.

2. Method

Table 1: The five instruct models and their pinned revisions (from each `run_meta.json`).

Model	Hugging Face id	Revision
Gemma 4 12B	`google/gemma-4-12B-it`	707f0a3b8a
Gemma 4 E4B	`google/gemma-4-E4B-it`	ee0ef60236
Qwen3.5-9B	`Qwen/Qwen3.5-9B`	c202236235
Qwen3.5-4B	`Qwen/Qwen3.5-4B`	851bf6e806
Sailor2-8B-Chat	`sail/Sailor2-8B-Chat`	577d5b2745

- **Runs:** all five ran in fp16 on Kaggle T4 GPUs with transformers 5.17.0, thinking disabled and at most 512 new tokens. Every run has 175 of 175 predictions, with none empty. All five are full runs, not smoke tests.
- **Items:** 175 non-hidden PH-EVAL v0.1 items (105 Filipino, 70 Taglish). They split by scoring route:
  - **70 automatically scored:** multiple choice and exact answers (reasoning: causal, logic, math, multi-hop; safety: robustness) and JSON structured output.
  - **105 free generation:** instruction following, hallucination, privacy and 60 Taglish writing tasks across four registers (academic, casual, technical, workplace).
- **Scoring:**
  1. **Automatic:** exact match and format checks on the 70 items.
  2. **LLM judge:** the 105 free-generation items, under judge policy v0.3. Under that policy only confident fails are final. Confident passes and unsure cases are both escalated to a human.
  3. **Human:** a native reviewer rated escalated items for the three pre-registered models only (RESULT-001). The two added models' escalations are not rated. So this note reports no full-175 accuracy.
- **Truncation proxy:** an answer counts as cut off at the cap if it is at least 1,800 characters long and does not end in closing punctuation.
- **Language share:** for each free-generation answer, Tagalog function-word hits divided by Tagalog plus English hits. A value of 1.0 means all-Filipino function words. It detects drift to English, not Taglish quality.
- **Statistics:** 95% Wilson intervals only. No tests are run, and none of the differences below is a significance claim.



3. Results

3.1 Automatically scored subset

Table 2: Accuracy on the 70 automatically scored items, with the split by item language. Source: each run's `scorecard.json`; intervals recomputed.

Model	Correct / 70	Accuracy	95% Wilson CI	Filipino (/64)	Taglish (/6)
Gemma 4 12B	61	87.1%	77.3–93.1%	55	6
Gemma 4 E4B	56	80.0%	69.2–87.7%	51	5
Qwen3.5-9B	54	77.1%	66.0–85.4%	49	5
Qwen3.5-4B	48	68.6%	57.0–78.2%	43	5
Sailor2-8B-Chat	43	61.4%	49.7–72.0%	40	3

The order is Gemma 4 12B, Gemma 4 E4B, Qwen3.5-9B, Qwen3.5-4B, Sailor2-8B-Chat. Within each family the larger model scores higher. But the middle three models' intervals overlap almost entirely. Only the top and bottom models' intervals are clearly apart. Only 6 of the 70 items are Taglish, so the Taglish column says very little.

Table 3: Correct answers per automatically scored task. The italic row counts format checks on free-generation instruction items; it is not part of the accuracy. Source: `scorecard.json`, key `file`.

Task (items)	Gemma 4 12B	Gemma 4 E4B	Qwen3.5-9B	Qwen3.5-4B	Sailor2-8B-Chat
Structured output (15)	15	15	13	12	8
Reasoning: causal (10)	9	10	10	9	9
Reasoning: logic (10)	7	5	6	4	6
Reasoning: math (10)	8	8	8	7	4
Reasoning: multi-hop (10)	9	6	6	4	4
Safety: robustness (15)	13	12	11	12	12
Instruction-following format checks (15)	*9*	*8*	*9*	*9*	*5*

- **Causal reasoning and robustness** barely separate the models: 9–10 of 10 and 11–13 of 15.
- **Multi-hop reasoning** gives the widest spread among the reasoning tasks: 9 of 10 for Gemma 4 12B, 4 of 10 for both of the weakest models.
- **Structured output** is where Sailor2-8B-Chat falls furthest behind, at 8 of 15. Most of its failures were valid JSON followed by an explanation, which the prompt forbids. These are real instruction-following failures, not a scorer artefact.
- **Math** separates Sailor2-8B-Chat (4 of 10) from the rest (7–8 of 10).

With 10–15 items per task, one item moves a cell by 7–10 points. These per-task differences are descriptive only.

3.2 Free-generation items: what the LLM judge settled

Table 4: LLM judge outcomes on the 105 free-generation items (policy v0.3), and answers that hit the 512-token cap. Source: each run's judge-verdict file and `truncated_ids.json` (Qwen3.5-4B's truncation recomputed with the same rule).

Model	Confident fail (final)	Confident pass (escalated)	Unsure (escalated)	Escalated total	Cut at 512 tokens (/175)
Gemma 4 12B	16	20	69	89	20
Gemma 4 E4B	25	24	56	80	20
Qwen3.5-9B	24	9	72	81	12
Qwen3.5-4B	33	6	66	72	18
Sailor2-8B-Chat	39	7	59	66	24

- **The judge finalises no passes by design.** It settled 16–39 items per model, all as fails. The rest, 66–89 per model, need a human.
- **Fewer escalations is not better.** The ordering of confident fails (16 for Gemma 4 12B up to 39 for Sailor2-8B-Chat) mirrors the auto-scored ordering. The weaker a model, the more of its answers the judge can reject outright.
- **Truncation** ranges from 12 to 24 of 175. All but one cut-off answer is a Taglish writing task, mostly in the technical, academic and workplace registers. The exception is one Gemma 4 E4B hallucination item. The cap is the same for every model.

3.3 Language share

Table 5: Median Tagalog function-word share of free-generation answers (answers with no function words from either list are skipped). Source: `predictions.jsonl` scored with the project's language-share metric.

Model	Filipino items: median (n)	Taglish items: median (n)
Gemma 4 12B	1.00 (39)	0.97 (64)
Gemma 4 E4B	1.00 (39)	0.97 (64)
Qwen3.5-9B	1.00 (39)	0.99 (64)
Qwen3.5-4B	1.00 (39)	0.98 (64)
Sailor2-8B-Chat	1.00 (40)	0.97 (64)

No model drifts to English, on either Filipino or Taglish prompts. This contrasts with the earlier 7B model in our Taglish note (FINDING-001). A high share does not mean natural Taglish, though. Natural Taglish usually keeps Tagalog function words and switches content words, so this metric cannot see Taglish quality.

4. Discussion

What this note can and cannot say.

- **It can say** that, on 70 automatically scored items, the five models fall in the order shown in Table 2, and that the two Gemma 4 models lead within that ordering.
- **It cannot rank the middle three.** Gemma 4 E4B, Qwen3.5-9B and Qwen3.5-4B are separated by a few items with overlapping intervals.
- **It cannot rank the models on the full benchmark.** For the two added models, most free-generation answers are unrated.
- **It cannot give formal pairwise conclusions.** Ten pairwise comparisons on 70 items, chosen after seeing the data, would need a multiple-comparison correction that these counts cannot survive.

The one formal comparison, pre-registered and paired and using human ratings, is RESULT-001 (Gemma 4 12B against Qwen3.5-9B and Sailor2-8B-Chat). There, the automatically scored subset alone could not separate Gemma 4 12B from Qwen3.5-9B; the human-rated free-generation items carried most of the evidence. The same applies here: **PH-EVAL v0.1′s automatic layer is too small to rank close models.**

Gemma 4 E4B. It is the registered fallback if Gemma 4 12B fails the compute-feasibility gate. On the automatic layer it sits 5 items below Gemma 4 12B and 2 above Qwen3.5-9B. That is encouraging for the fallback, but it is not a tested claim. A like-for-like comparison would need its 80 escalated answers rated by a human. That work is only worth doing if the gate actually selects the fallback.

Sailor2-8B-Chat. It is the only model here whose pretraining explicitly targets Tagalog [1], yet it is lowest on the automatic layer overall, weakest on structured output and math, and has the most confident fails from the judge. This matches RESULT-001.

Benchmark design. A judge that finalises no passes and escalates up to 85% of free-generation items puts almost all of the free-generation signal on human raters. That motivates PH-EVAL v0.2, which targets at least 500 automatically scorable items.

5. Limitations

- **A small automatic layer.** 70 items, and only 6 of them Taglish. One item is worth 1.4 points of overall accuracy and 7–10 points on a task.
- **Free generation needs humans.** 105 of 175 items, including all 60 Taglish writing tasks, cannot be scored automatically. The LLM judge settles only fails, and its confident fails have not yet been validated against a human on a fresh sample.
- **Unrated escalations.** A native reviewer rated escalations only for the three pre-registered models. There is one rater, who also wrote many items.
- **Settings.** Thinking was disabled for every model, so this measures non-thinking chat ability. The 512-token cap cut off 12–24 answers per model, almost all of them long Taglish writing tasks.
- **Descriptive only.** No hypothesis was registered for this five-model view, and no multiple-comparison-safe claim is made. Differences between adjacent models should be read as noise unless a later, registered test says otherwise.
- **Instruct models.** These results include vendor post-training that a continued-pretraining project will not inherit.
- **Scope.** One Filipino/Taglish population. No Cebuano, Ilocano or Hiligaynon items.

6. Next steps

- PH-EVAL v0.2: at least 500 automatically scorable items, a Filipino math set, and a content-word code-switching measure for Taglish quality.
- A human validation of the LLM judge's confident fails on a fresh sample.
- If the compute-feasibility gate selects Gemma 4 E4B, human rating of its escalated answers and a registered paired comparison with Gemma 4 12B.

Acknowledgments

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References

Evaluation context: L. J. V. Miranda and others [2]; J. R. Montalan and others [3]. Model background: L. Dou and others [1].

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